

BILLUND AP, Denmark

WMO: 061040

Lat: 55.738N

Lon: 9.168E

Elev: 247

StdP: 14.57

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	15.4	19.2	11.4	9.9	18.7	14.9	11.8	22.0	31.8	43.1	28.5	43.1	3.4	120	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.7	80.3	65.0	76.8	63.6	73.3	61.8	67.1	75.4	65.3	73.0	63.6	70.7	9.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.5	69.9	62.6	85.6	68.1	60.9	80.4	66.0	31.7	75.3	30.2	73.0	29.0	70.7	74.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 24.6	21.6	19.3	DB	7.8	84.8	5.5	2.9	3.8	86.9	0.6	88.7	-2.5	90.3	-6.6	92.4	(4)
(5)			WB	7.3	70.3	5.4	2.0	3.4	71.7	0.2	72.8	-2.8	73.9	-6.7	75.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	47.0	33.7	34.0	37.6	45.2	52.0	57.0	61.6	61.5	55.7	48.4	40.9	36.0		
	DBStd	11.79	7.09	6.92	6.26	6.23	5.98	5.13	5.29	5.14	4.84	5.76	6.22	7.55		
	HDD50	2374	506	447	385	169	47	3	0	0	7	98	277	435		
	HDD65	6624	971	867	849	594	404	246	137	137	281	515	723	900		
	CDD50	1295	0	0	0	26	109	214	360	356	179	47	4	0		
	CDD65	70	0	0	0	0	2	7	32	27	2	0	0	0		
	CDH74	599	0	0	0	2	18	79	284	206	10	0	0	0		
	CDH80	88	0	0	0	0	1	9	46	32	0	0	0	0		
	WSAvg	9.1	10.2	10.4	10.3	9.3	9.1	9.1	8.1	7.8	8.2	8.7	8.8	9.2		
	PrecAvg	36.4	3.4	2.5	2.2	1.8	2.1	2.8	3.0	3.7	3.6	4.0	3.5	3.8		
(15) Precipitation	PrecMax	44.7	6.5	5.6	4.9	3.7	4.1	5.8	5.3	7.3	7.2	9.9	7.0	7.7		
	PrecMin	24.3	0.2	0.8	0.4	0.2	0.2	0.0	0.7	1.1	0.7	1.5	1.0	1.0		
	PrecStd	5.3	1.6	1.2	1.1	0.8	1.0	1.4	1.3	1.7	1.7	1.7	1.5	1.9		
	PrecStd	5.3	1.6	1.2	1.1	0.8	1.0	1.4	1.3	1.7	1.7	1.7	1.5	1.9		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.6	49.6	57.4	71.2	76.6	80.9	84.5	83.7	75.4	64.0	56.9	51.7		
		MCWB	46.9	46.8	47.0	56.6	61.7	65.0	66.7	66.1	63.0	58.6	54.6	50.3		
	2%	DB	46.5	46.7	52.2	64.7	71.8	75.9	80.7	79.5	70.2	60.5	53.7	49.4		
		MCWB	45.2	45.0	45.4	52.7	58.4	62.6	65.2	65.0	60.7	56.6	51.8	48.3		
	5%	DB	44.7	44.9	48.5	60.5	67.9	71.7	77.1	75.4	66.5	58.5	51.6	47.6		
		MCWB	43.4	43.5	44.0	50.5	56.4	60.4	64.0	63.5	59.0	55.7	49.8	46.6		
	10%	DB	43.0	43.1	46.1	55.8	64.2	67.8	73.2	71.5	64.0	56.6	49.6	45.9		
		MCWB	41.8	41.8	43.1	48.3	54.5	58.8	62.4	62.0	58.2	54.2	48.2	44.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	47.3	47.1	49.5	57.8	63.6	67.5	69.9	69.3	65.2	60.1	55.1	50.6		
		MCDB	48.1	47.9	53.1	68.4	73.6	76.0	79.1	77.2	71.6	62.8	56.5	51.3		
	2%	WB	45.3	45.3	46.8	54.3	60.6	64.5	67.5	66.8	62.3	57.9	52.2	48.5		
		MCDB	46.1	46.2	49.9	62.6	68.3	72.9	76.4	74.4	67.5	59.6	53.5	49.2		
	5%	WB	43.4	43.7	44.9	51.6	57.9	62.0	65.4	65.1	60.7	56.1	50.1	46.5		
		MCDB	44.3	44.5	47.3	58.2	65.1	69.2	73.4	72.5	64.8	57.9	51.2	47.1		
	10%	WB	41.7	41.9	43.3	49.6	55.6	59.7	63.6	63.5	59.3	54.5	48.1	44.7		
		MCDB	42.6	42.9	45.4	54.6	62.7	65.8	70.9	70.0	62.8	56.3	49.0	45.5		
(35) Mean Daily Temperature Range	5% DB	MDBR	7.9	8.6	11.9	15.8	17.1	15.9	16.7	16.4	13.9	11.2	8.7	7.9		
		MCDBR	7.5	8.6	16.1	22.2	23.7	23.1	23.2	21.9	17.6	12.2	8.4	7.1		
		MCWBR	7.7	7.4	10.5	12.1	12.6	12.1	10.5	9.9	9.9	9.2	7.9	7.3		
		MCDBR	7.4	7.9	13.1	19.4	20.2	19.8	19.8	18.3	14.4	10.6	8.3	7.2		
	5% WB	MCWBR	7.6	7.3	9.7	11.6	12.1	11.8	10.3	9.5	9.1	8.8	8.2	7.4		
(40) Clear-Sky Solar Irradiance	taub		0.318	0.337	0.360	0.394	0.386	0.381	0.396	0.397	0.374	0.353	0.331	0.304		
	taud		2.239	2.282	2.283	2.241	2.315	2.352	2.327	2.336	2.385	2.381	2.298	2.232		
	Ebn at Noon		183	229	253	261	271	274	266	258	247	221	178	158		
	Edh at Noon		18	25	31	38	37	36	37	34	29	23	17	14		
(44) All-Sky Solar Radiation	RadAvg		154	346	767	1279	1673	1776	1649	1316	909	469	190	112		
	RadStd		27	48	71	152	163	140	177	97	94	56	29	12		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	+0.65	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (45 neighbors)	+0.67	N/A	N/A	N/A	N/A	N/A	-211	-293	N/A	N/A

Nomenclature: See separate page